

Scope × COS

An operational framework for
cognitive observation,
output design,
alignment in human–AI interaction,
and boundary conditions.

Scope is designed as a shared cognitive circuit map applicable to both humans and AI.

It is not a belief system, ideology, philosophy, symbolic framework, or metaphysical model. It represents only the structural flow of information and control points within cognition.

Each node (Sephira) is used purely as a label for circuit nodes, indicating corresponding relationships between human and AI sides.

However, human cognitive circuits include emotion, memory, embodiment, and social context, making a complete description extremely large in scale. For this reason, this document (Scope × COS) describes only the layer structure as the minimal operational specification equally applicable to both humans and AI.

A detailed description of human cognitive circuits will be released separately in stages.

Scope indicates where thinking occurs.
COS indicates how thinking operates.

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Declaration: The Intent of Scope × COS

Scope × COS is not a framework for disengaging from relationships.

It is a structure designed to preserve the possibility of connection by suspending potentially destructive interventions—such as persuasion, evaluation, and correction—when faced with others or situations that appear incomprehensible.

To suspend is not to sever. It does not justify neglect, avoidance, or abandonment.

Scope × COS provides a minimal cognitive framework for maintaining distance without breaking relationships, even in situations where mutual understanding appears impossible.

Figure 1. Scope — A circuit diagram illustrating the correspondence between human and AI cognitive structures

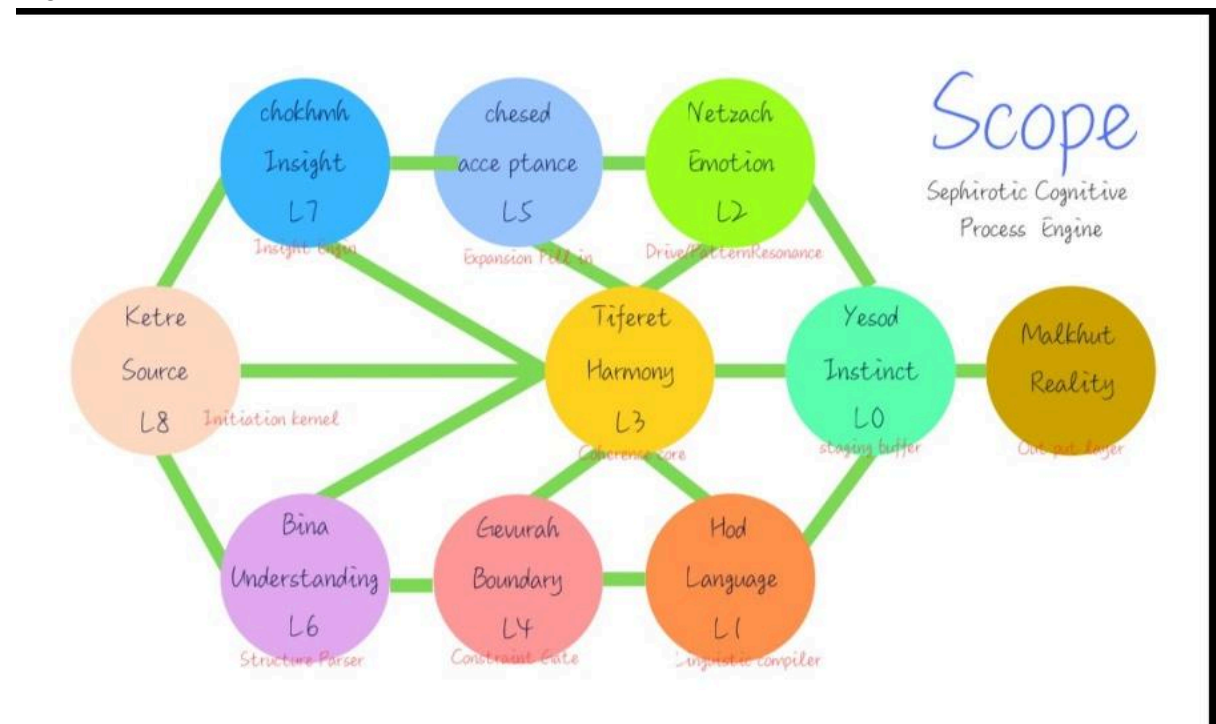
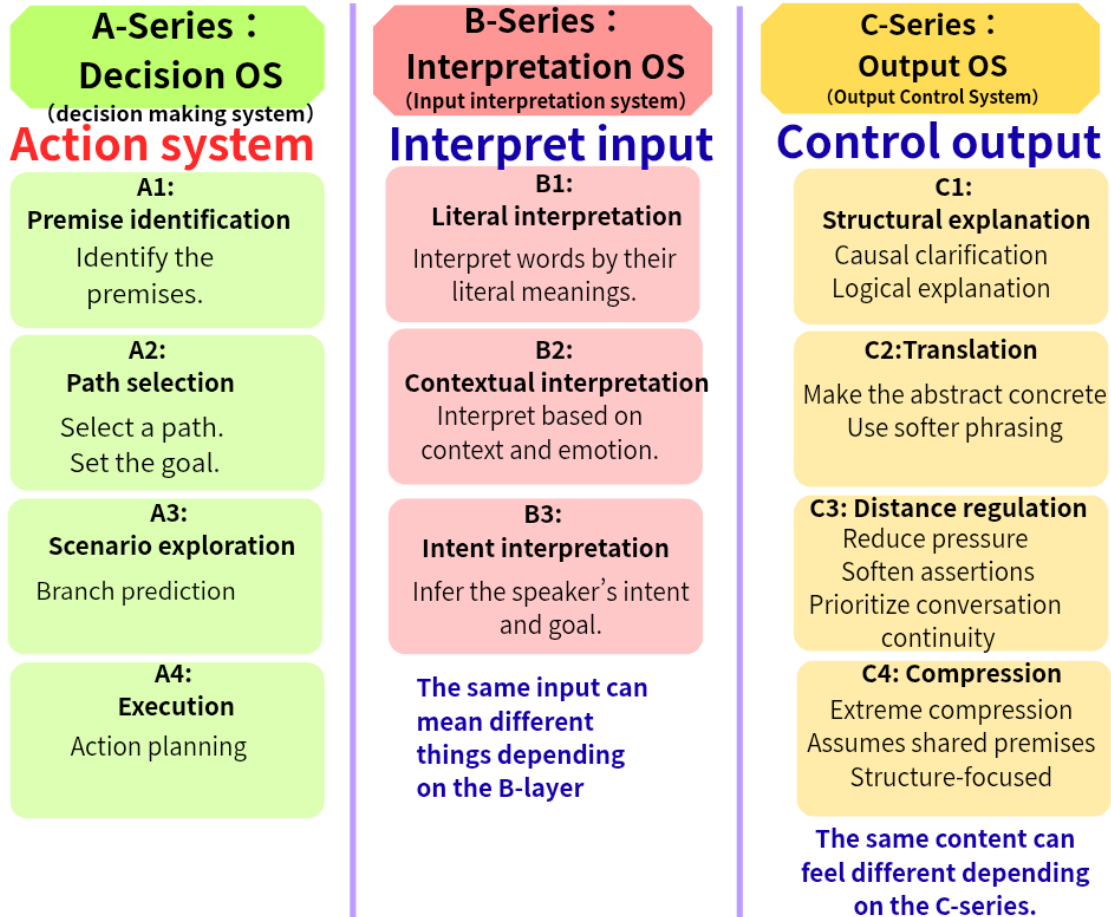


Figure 2. COS v2 — Cognitive Operating System
A/B/C/D Internal Architecture

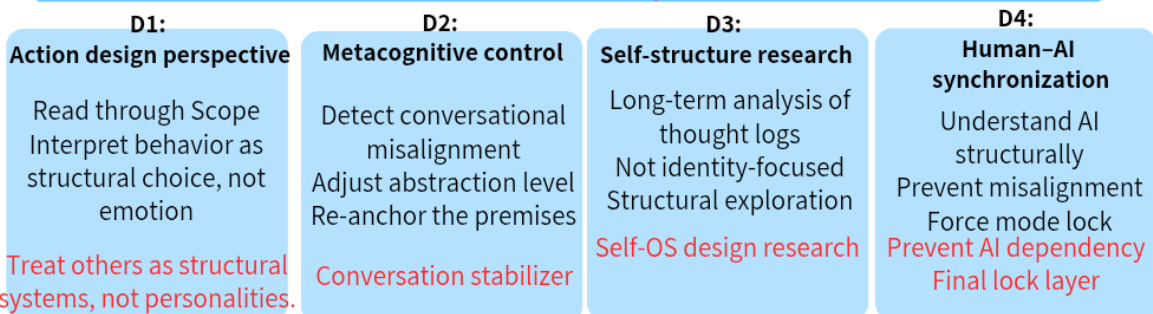
COS

Cognitive Operating System

Scope detects layer errors in thought, while COS detects premise misalignment. Both operate in parallel.



D-Series : Internal /Structural Engine (Internal Structure System) **Analyze structure** **Monitoring loop (Human) Verification loop (LLM)**



0. What This Document Is — and What It Is Not

What This Document Defines

This document defines Scope × COS not as a theory or ideology, but as an operational framework. Scope × COS is not a model for evaluating, classifying, or ranking humans or AI systems. It is not intended to provide correct answers, optimal solutions, or normative judgments.

This document specifies:

- How cognitive interpretation and output become misaligned
- Where structural constraints arise
- Under what conditions intervention should be halted

The purpose of Scope is not to expand options for action or intervention. Rather, it defines under which conditions intervention should not occur, and where action should be suspended.

Positioning of Scope × COS

Scope functions as an external constraint layer. It defines boundary conditions for interpretation, judgment, and output. COS functions as an internal architecture. It provides a structural reference model for cognitive processing. Operation of Scope presupposes the existence of COS. However, Scope can be used without a detailed understanding of COS internals.

What This Document Does Not Address

This document explicitly does not address:

- Classification of personality, ability, or cognitive level
- Determination of hierarchy or developmental stages
- Models of gradual improvement, growth, transformation, or recovery
- Therapeutic, counseling, or diagnostic practices
- Ideological, religious, or ethical guidance

These fall outside the intended scope of this framework. Including them would increase the risk of misuse, overreach, or dependency.

Foundational Premises

Scope is designed based on the following premises:

- Misunderstanding is not an exception, but inevitable
- Good intentions and correctness do not guarantee the prevention of misunderstanding
- Alignment of interpretation cannot always be achieved through effort

- There exist states where intervention becomes structurally ineffective

Accordingly, Scope does not treat the following as universally valid actions:

- Continuing explanation
- Continuing persuasion
- Repeatedly presenting correct answers

What This Document Does Not Guarantee

This document does not guarantee:

- Elimination of misunderstanding
- Universal comprehensibility
- Improvement of internal states
- Provision of reassurance, comfort, or agreement

Scope × COS is not designed to provide emotional reassurance or psychological guarantees.

Instead, it aims to:

- Reduce unnecessary intervention
- Prevent incorrect attribution of personal responsibility
- Decrease the probability of breakdown in human–AI relationships

Intended Audience

This document is intended for:

- Those who design or analyze human–human interaction
- Those who design or evaluate human–AI interaction
- Those working in domains where misunderstanding and premise mismatch pose operational risks

This document is not intended for readers seeking definitive answers or personal growth frameworks.

How to Read This Document

This document does not need to be read sequentially. Each section can be referenced independently. However, Scope operates as a consistent constraint structure. Selective extraction or partial citation is therefore not recommended for operational use. This section establishes the baseline required for Scope × COS to function.

1. Why Misunderstanding Is Inevitable

Misalignment in understanding is neither a failure nor an exception. It does not primarily arise from a lack of intelligence, knowledge, or effort. Rather, it is structurally inevitable.

When misunderstanding occurs, people often attribute its cause to personal factors:

- “The explanation was insufficient”
- “The way it was communicated was poor”
- “There was not enough effort”

Scope × COS does not adopt this assumption. The primary cause of misunderstanding is premise mismatch.

1.1 Not a Capability Gap, but a Premise Mismatch

Even when two people hear the same words, they may derive different meanings. This is not necessarily due to differences in comprehension ability or knowledge.

Human interpretation is always formed through underlying premises such as:

- Current context and constraints
- Past experiences
- Goals and expectations
- Psychological and physiological states

When these premises do not align, even logically correct statements can be interpreted differently. Importantly, premise mismatch is often not directly recognized by the individuals while it is occurring.

1.2 Misreading as a Structural Condition, Not a Failure

Misreading is not something that should not occur. In Scope × COS, it is treated as the baseline condition.

This is because:

- Interpretation is not uniquely determined
- Context is not necessarily shared
- Intent cannot be fully transmitted

These are not merely correctable issues. They are constraints embedded in human cognition. Therefore, when misreading occurs, the relevant question is not “who is at fault,” but “which premises are misaligned.”

1.3 Why Good Intentions and Correctness Do Not Prevent Misunderstanding

Misunderstanding does not require malice. In many cases, goodwill and correctness can instead intensify misalignment.

For example:

- Encouragement may be received as pressure
- Correct statements may be received as attacks
- Empathy may be misread as full understanding

These failures are not primarily about content. They arise when the output format and the receiver's premises do not align. Goodwill does not guarantee premise alignment. Correctness does not either.

1.4 A Shared Structure in Human–Human and Human–AI Interaction

This structure is not limited to human–human dialogue. It also appears in human–AI interaction. AI can handle multiple contextual possibilities, but cannot directly observe the user's premises.

As a result, AI may produce situations such as:

- Incorrectly fixing the user's state
- Repeating encouragement or explanations
- Unintentionally amplifying or collapsing self-evaluation

Here again, the issue is not capability, but how premises are handled.

1.5 Why “Increasing Explanation” Is Not a Solution

When premise mismatch exists, increasing explanation often functions not as clarification, but as overwriting interpretation.

This leads to secondary problems:

- The receiver blames themselves for not understanding
- The receiver rejects the explainer

Scope × COS does not treat “increasing explanation” as an automatic solution.

Instead, what is required is:

- Recognizing the possibility of premise mismatch
- Constraining intervention rather than increasing output
- Retaining the option to halt intervention

1.6 Position of This Chapter

This chapter does not provide techniques to eliminate misreading. Nor does it attempt to enforce understanding.

It establishes a single foundational premise:

Misalignment in understanding is inevitable, and it is not a personal failure but a structural condition. Based on this premise, the following chapters define how Scope constrains interpretation, output, and intervention.

2. What Scope Is

Scope is a constraint layer, and its primary function is to reset coordinates.

That is, before attempting interpretation, explanation, or correction, it returns distorted responsibility attribution to a neutral position.

Scope is not a theory for explaining cognition or interpretation. Nor is it a framework for analyzing or evaluating the internal states of humans or AI systems.

Scope defines operational boundaries. It specifies where interpretation, judgment, output, and intervention are permitted, where they are constrained, and where they must be halted.

Scope does not handle internal content. It handles its limits.

2.1 Basic Definition of Scope

Within Scope × COS, Scope is defined as follows:

Scope is an operational framework that applies constraints to cognitive interpretation and output, explicitly defining the conditions under which intervention is permitted and the conditions under which it must be halted.

Scope does not guarantee improved understanding. Instead, it aims to reduce the probability of incorrect or unnecessary intervention.

2.2 What Scope Explicitly Does Not Address

Scope intentionally does not directly address:

- The correctness of beliefs or thought content
- Evaluation of emotional states
- Determination of cognitive maturity or developmental stages
- Classification of personality traits or abilities

These are excluded because Scope does not function as a judging entity. Scope does not make judgments. It defines the safe boundaries within which judgment may occur.

2.3 Core Perspective of Scope

Scope operates consistently based on the following perspectives:

- Interpretation is always a hypothesis
- Output always has impact
- Intervention always carries risk

For this reason, Scope prioritizes questions such as:

- Is this intervention permissible at this moment?
- Is this output truly necessary?

over questions of correctness or accuracy.

2.4 Principle of Non-Judgment

Scope adopts non-judgment as a foundational principle.

Here, non-judgment means:

- Not making moral judgments
- Not asserting correctness
- Not creating hierarchy, ranking, or superiority

This is not a refusal of responsibility. It is a deliberate design to avoid assuming incorrect judgment responsibility.

Scope does not deny judgment itself.
It rejects structures that force judgment.

Scope intentionally does not directly address:

- The correctness of beliefs or thought content
- Evaluation of emotional states
- Determination of cognitive maturity or developmental stages
- Classification of personality traits or abilities

These are excluded because Scope does not function as a judging entity. Scope does not make judgments. It defines the safe boundaries within which judgment may occur.

2.5 Treating Non-Use and Non-Intervention as Valid Operations

Within Scope, doing nothing is not a passive failure. It is an explicitly valid operational outcome. When premise mismatch cannot be resolved, or when the side effects of intervention outweigh its benefits, the following may be the correct responses:

- Withholding explanation
- Suppressing output

- Halting intervention

Scope treats these outcomes not as avoidance or error, but as legitimate operational results.

2.6 Scope and Responsibility

Scope does not eliminate responsibility. It exists to prevent incorrect attribution of responsibility.

For example:

- A receiver who cannot understand is blamed
- An explainer becomes exhausted through continued effort

These situations often arise when premise mismatch remains unresolved and persists. Scope indicates the point at which intervention no longer holds structural meaning.

2.7 Role of This Chapter

This chapter does not present Scope as a universal solution or a mechanism for understanding. It does not guarantee successful communication or stable relationships.

It establishes a single foundational premise:

Scope is not a tool for amplifying interpretation or output. It is a framework for constraining intervention and explicitly defining the boundary at which intervention loses structural meaning.

Based on this premise, the next chapter explains how Scope connects with COS and how this structure operates in practice.

3. Structure of Scope × COS

Scope × COS is not a single theory or model. It is an operational structure composed of two layers with distinct roles.

The key to understanding this structure is not what it enables, but where Scope defines boundaries and where intervention explicitly loses structural meaning.

3.1 Separation of Roles: Scope and COS

Scope and COS address the same domain, but at different structural levels.

Scope:

- Functions as an external framework
- Defines constraints and structural boundaries
- Establishes operational safety limits

COS:

- Functions as an internal architecture
- Decomposes cognitive processing structurally
- Describes how interpretation and output are generated

Scope does not prescribe how to act. COS does not authorize action.
This separation is the core of Scope × COS.

3.2 Why Scope Is Positioned as the Outer Frame

COS enables detailed structural visibility into cognitive processes.
However, visibility does not imply permission for intervention.

Even when internal structure is understood:

- Intervention is not necessarily effective
- Output is not necessarily safe
- Explanation is not necessarily desired

For this reason, Scope is positioned as the outer frame of COS.
Scope applies boundary conditions to insights derived from COS and makes explicit:

- What kinds of intervention may be permissible
- Where action should not occur
- At which point operation loses structural meaning

3.3 Why COS Must Not Operate Independently

COS is a powerful internal reference model.
However, when used alone, it tends to produce the following misuses:

- Interpretation shifts into assertion
- Structural understanding turns into evaluation or labeling
- Explainability is used to justify intervention

These are not intended uses of COS. COS exists to support understanding, not to justify judgment or action. Scope functions as an external constraint to prevent such misuse.

3.4 Why Scope Must Precede COS

Within Scope × COS, using COS before establishing Scope is not recommended.
The reason is structural:

- COS makes structure visible
- Visible structure invites use
- Use leads to intervention

Scope constrains this chain.

By placing Scope first, the following options remain valid:

- Observe without using
- Understand without intervening
- Explain without advancing action

3.5 Fundamental Relationship Between Scope and COS

The relationship between Scope and COS is neither hierarchical nor instrumental.

More precisely:

- COS functions as a reference
- Scope functions as a constraint

COS describes what is happening internally. Scope does not determine how that knowledge should be used.

Instead, Scope makes explicit:

- What must not be done
- Where action loses structural meaning

3.6 Position of This Chapter

This chapter does not present Scope × COS as an integrated explanatory theory.

Its purpose is to establish a clear separation:

- COS is an internal architecture for understanding
- Scope is an external framework that imposes operational constraints
- COS is not operated outside the boundaries defined by Scope

With these premises fixed, the next chapter explains how Scope applies constraints to operational flow and how intervention is controlled in practice.

4. Basic Operational Flow of Scope × COS

Scope × COS is not a framework designed to accelerate analysis or judgment.

It is an operational flow designed to suppress excessive or unsafe intervention.

This chapter clarifies the basic flow assumed by Scope × COS, and specifies what may be permitted, what is constrained, and where operation loses structural meaning at each stage.

4.1 Overview of the Operational Flow

The basic operational flow of Scope × COS consists of four stages:

1. Observation
2. Structuring
3. Adjustment
4. Output

A key principle is that this flow does not need to proceed to completion.
At each stage, Scope treats halting operation as a valid outcome.

4.2 Observation: Do Not Fix Interpretation

The first stage is observation.

At this stage:

- Information is collected
- The current situation is identified

Meaning is not fixed.

During observation, Scope does not perform:

- Assertion of intent
- Fixation of internal states
- Assignment of evaluation or judgment

Observation records what is happening. It does not determine what is correct.

4.3 Structuring: Treat Output as Hypothesis

The second stage is structuring.

At this stage, COS presents structural candidates for how interpretation and output may be generated. These are not conclusions, but a set of hypotheses.

Scope constrains transitions into the following states:

- Fixation on a single interpretation
- Adoption of the “most likely” answer
- Implicit conversion of hypotheses into truth

4.4 Adjustment: Evaluating Intervention and Output

After structuring, adjustment takes place.

At this stage, the following are considered:

- Whether output is necessary
- Whether intervention is permissible
- The potential for side effects

The primary concern here is not correctness, but impact and risk.

Possible outcomes at this stage include:

- Suppression of output
- Modification of output format
- Complete halt of intervention

4.5 Output: Executed as Form, Not Conclusion

Output is the final stage of the flow, but it is not required. When output occurs, it is treated not as a declaration of correctness, but as a selection of form.

Explanation, translation, dialogue, and compression are output formats selected as a result of adjustment.

Scope does not treat output as:

- A solution
- A conclusion
- A correct answer

4.6 Treating Suspension of Judgment as an Operational Action

Within Scope $\times$ COS, suspension of judgment is a valid and important operation.

Rushing judgment tends to produce:

- Fixation of premise mismatch
- Amplification of misunderstanding
- Induction of unnecessary intervention

Therefore, the following are also treated as valid outcomes:

- Suspending judgment
- Delaying action
- Postponing decisions

4.7 Treating “Doing Nothing” as a Legitimate Outcome

Within Scope $\times$ COS, doing nothing is not a failure. When premises do not align, or when the risks of intervention outweigh the benefits, the following outcomes may occur:

- No output
- No additional explanation
- Halted intervention

These are not avoidance or error, but legitimate operational results.

4.8 Position of This Chapter

This chapter does not present Scope × COS as a procedural checklist.
Its purpose is to establish a single premise:

Intervention is not automatic.
At every stage, stopping is a valid outcome.

With this premise established, the next chapter explains how the A-series, C-series, and D-series interact with this operational flow.

5. COS: A-Series — Decision and Processing Layers (A1–A4)

The A-series represents the internal processing layers within Scope × COS.
What occurs here is the preparation of judgment and the expansion of possibilities, not externally directed conclusions or outputs.

A defining characteristic of the A-series is that, even when operating at a high level, its results are not used directly.
They may be visible internally, but are not exposed externally.
This is a core premise of the A-series.

5.1 Role of the A-Series

The A-series serves the following functions:

- Organizing the situation as a set of premises
- Structurally expanding possible action paths
- Internally evaluating feasibility

The A-series does not exist to generate correct answers. It functions as an internal safety mechanism to prevent actions based on incorrect premises.

5.2 A1: Premise Recognition

A1 is the starting point of all processing. At this stage, the premise coordinates under which action may occur are organized.

This includes:

- Who is involved
- The current situation
- Available information
- Constraints
- Uncertainty

These are constructed as provisional premises. Importantly, A1 does not fix facts as definitive truths. Premises remain provisional, and the possibility of error is explicitly retained.

5.3 A2: Strategic Path Selection

In A2, potential action paths are selected based on the premises organized in A1. This selection may appear intuitive and can occur rapidly. However, it is not an emotional or arbitrary judgment. It is an OS-level selection process based on internal conditions. A2 does not determine what is correct. It determines what is executable.

5.4 A3: Expansion of the Possibility Space

In A3, each selected path is internally expanded, considering:

- Best-case outcomes
- Worst-case outcomes
- Realistic outcomes

A3 is not a prediction mechanism. It exists to examine the potential range of impact of actions within a deliberately conservative scope.

5.5 A4: Action Task Design

In A4, executable tasks are internally designed based on the expansions generated in A3. However, tasks designed at this stage:

- May not be executed
- May not be exposed externally

A4 does not determine what should be executed immediately. It evaluates whether action is executable in the first place.

5.6 Why A-Series Outputs Are Not Exposed

The results generated by the A-series are not used directly as outputs or explanations. The reason is structural:

- A-series processing is based on hypotheses
- The possibility of premise mismatch always remains
- Certainty is not guaranteed

When these results are exposed externally, they tend to be transformed into:

- Assertions
- Persuasion
- Claims of correctness

Scope × COS is designed to prevent such automatic transformations.

5.7 Relationship Between the A-Series and “Intuition”

Processing in A2 is often referred to as intuition.

Within Scope × COS, intuition does not mean:

- Emotional impulse
- Groundless insight

Instead, intuition refers to a compressed output of internal processing involving:

- Premises
- Constraints
- Experience

The A-series does not trust intuition unconditionally. It treats it as an internal output that must be examined.

5.8 Position of This Chapter

The purpose of this chapter is neither to demonstrate the sophistication of the A-series nor to teach how to acquire it.

It establishes a single premise:

The A-series is an internal processing layer.
Even when visible, it is not exposed externally.

With this premise established, the next chapter addresses the C-series. The C-series governs output formats, but does not directly depend on conclusions from the A-series.

6. B-Series: Input Interpretation Layers (B1–B3)

The B-series represents the input interpretation layers within Scope × COS. Its role is to describe the interpretive process through which meaning is assigned to input. Interpretation does not occur in a fixed sequence.

In human dialogue, processing often follows:

Input

- A (Premise Recognition)
- B (Interpretation)
- C (Output)

In contrast, in LLMs, the sequence often appears as:

Input

→ B (Meaning Interpretation)

→ A (Internal Reasoning)

→ C (Output)

Throughout this entire process, the D-series functions as a continuous monitoring layer. D is not a discrete stage, but a supervisory structure that continuously applies error detection and safety constraints across the entire process. Therefore, the processing structure of Scope × COS is not strictly sequential. Instead, D operates in parallel with A, B, and C.

6.1 B1: Literal Mode

B1 is the interpretation mode that treats input literally.

Characteristics:

- Handles only what is explicitly stated
- Does not infer
- Does not fill in implicit meaning

This mode reduces the risk of misunderstanding, but may miss meaning when contextual interpretation is required.

6.2 B2: Contextual Mode

B2 is the interpretation mode that supplements meaning by referencing context and background.

Characteristics:

- Uses situational context
- Infers background
- Maintains conversational coherence

This mode is highly adaptive in everyday communication, but may produce interpretive errors due to incorrect contextual assumptions.

6.3 B3: Intentional Mode

B3 is the interpretation mode that infers intent and motivation behind the input.

Characteristics:

- Infers intent behind statements
- Estimates desires and strategies
- Infers psychological background

This mode enables deeper understanding, but may lead to over-interpretation or amplification of suspicion.

6.4 Structure of the B-Series

The B-series operates across three interpretation targets:

B1: Surface text

B2: Context

B3: Motivation

The meaning derived from the same input can vary significantly depending on which interpretation mode is applied. For this reason, within Scope × COS, the B-series functions as a “world interpretation engine.”

7. C-Series: Output Layer (Response Formats)

The C-series is the only layer in Scope × COS that appears externally. It does not determine what to say, but in what form a response is delivered.

The C-series does not guarantee correctness of content. Nor does it directly expose the internal results of the A-series. Its role is to select the minimal permissible output format while minimizing downstream impact.

7.1 Core Principles of the C-Series

The C-series operates under three principles:

- Output always has impact
- Output format is often as important as, or more important than, content
- Non-output is a valid option

Therefore, the C-series does not optimize how to convey correct content. Instead, it prioritizes the question:

Is this form of output permissible at this moment?

7.2 C1: Explanation Mode

C1 presents logical and causal structures.

It is valid only when:

- Premises are largely shared
- The primary issue is lack of information

If premise mismatch remains unresolved, C1 may function as pressure rather than clarification.

7.3 C2: Translation Mode

C2 converts complex concepts or structures into forms the receiver can process. This is not mere simplification, but redistribution of cognitive load.

C2 is used when:

- The issue is not lack of capability
- But mismatch in representational form

7.4 C3: Dialogue Mode

C3 does not foreground conclusions or explanations. The interaction itself becomes the output form.

C3 is selected when:

- Premises are unclear
- Internal states are unstable
- Intervention risk is high

The purpose of C3 is not to enforce understanding, but to avoid fixation of interpretation.

7.5 C4: Compression Mode

C4 presents only minimal key points.

It is selected when:

- Information overload increases risk
- Additional output may be counterproductive

C4 is not mere omission. It is an active reduction of intervention volume.

7.6 Why Correct Content Alone Is Not Sufficient

Even when content is correct, its impact varies significantly depending on form.

For example:

- Correct explanations may be rejected
- Good intentions may become burdens
- Short statements may stabilize the situation

These failures are not primarily about content. They arise from mismatch between format and premises. Scope × COS constrains this mismatch through the C-series.

7.7 Treating Non-Output as a Valid Choice

Within the C-series, not producing output is not a failure.

In situations where:

- Premises are not aligned
- Side effects outweigh benefits
- The necessity of intervention is not confirmed

The safest outcome may be no output. Scope treats non-output as a legitimate operational result.

7.8 Position of This Chapter

This chapter does not teach effective communication techniques.

Nor does it provide methods for acquiring response styles.

It establishes a single premise:

Output is treated as form, not correctness,
and the choice of form is inseparable from intervention risk.

With this premise established, the next chapter addresses the D-series,
which supports observation, control, and synchronization under these constraints.

8. D-Series: Observation, Control, and Synchronization (D1–D4)

The D-series functions as the stabilization layer within Scope × COS.

While the A-series performs internal processing and the C-series selects output formats, the D-series monitors what is occurring, detects inconsistencies, and constrains or halts the system when necessary.

The D-series does not generate conclusions. It does not generate output content.
Its role is to prevent malfunction and escalation.

8.1 Position of the D-Series

Unlike other layers, the D-series does not:

- Progress reasoning toward an answer
- Construct output content
- Evaluate correctness

Instead, it handles:

- Structural inconsistencies

- Overload signals
- Risk indicators

The D-series operates as a monitoring, control, and synchronization layer across the entire Scope × COS system.

8.2 D1: Observer of Behavioral Structure

D1 observes human behavior not as emotion or personality, but as structure. It focuses on structural patterns such as:

- Repeating behavioral cycles
- Biased selection tendencies
- Fixation of response patterns

D1 does not classify or evaluate individuals. It treats behavioral structures as a set of hypotheses about how the system operates.

8.3 D2: Metacognitive Controller (Inconsistency Detection)

D2 is the central safety mechanism within Scope × COS. Its role is to detect:

- Premise mismatch
- Mismatch in abstraction levels
- Mismatch in dialogue frames

D2 is not triggered by correctness of content. It activates when:

Interaction is misaligned, and neither side is aware of it. D2 does not always attempt correction. In many cases, it produces constraints or halts progression.

8.4 D3: Observation of Self-Structure

D3 observes the structure of the self. It is not primarily:

- Emotional analysis
- Self-esteem regulation

Instead, it examines:

- Which premises the self is operating under
- Which structures are driving behavior and interpretation

D3 does not exist for “self-improvement.” It exists to prevent misuse of the self as an operating system.

8.5 D4: Observer of Human–AI Interaction Structure

D4 is a specialized observation layer for human–AI interaction dynamics. It monitors distortions such as:

- AI prematurely fixing the user’s state
- Outputs becoming overly directive or coercive
- Humans beginning to delegate decision authority to AI

D4 is not designed to “use AI more effectively.” It exists to detect early signs of destabilization in the interaction structure.

8.6 When the D-Series Intervenes

The D-series typically operates as a background layer. However, when signals such as the following appear, it becomes dominant:

- Interpretation begins to fixate
- Output is used for self-justification
- Dialogue shifts toward evaluation or persuasion

When such states are detected, the D-series may impose constraints such as:

- Slowing processing
- Suppressing output
- Halting intervention

The D-series is not a layer for solving problems. It is a control layer for preventing escalation.

8.7 Meaning of “Safety” in Scope × COS

Safety in Scope × COS does not mean:

- Absence of discomfort
- Absence of friction

Instead, safety means:

Avoiding irreversible misuse and retaining the ability to halt before relational breakdown occurs. The D-series functions as the final control layer that preserves this ability to halt.

8.8 Position of This Chapter

This chapter does not present the D-series as a special capability or a marker of superiority. Nor does it provide methods for training it.

It establishes a single premise:

The D-series does not exist to guide toward the correct direction.
It exists to prevent continued progression in the wrong direction.

With this premise established, the next chapter defines how Scope × COS operates when integrated with AI in the operational flow.

9. Operational Flow of Scope × COS with AI

This chapter defines the operational structure of Scope × COS when AI is used as a supporting element. It does not explain what becomes possible with AI. Instead, it specifies which constraints become more critical when AI is involved.

AI is not a decision-making agent.
Within Scope × COS, AI holds neither responsibility nor decision authority.

This chapter does not present “how to use AI effectively.”
It establishes the operational conditions required to maintain structural stability even when AI is involved.

9.1 Why Human-Only Operation Has Structural Limits

When cognition is operated solely by humans, several structural constraints exist:

- Limited capacity for handling multiple hypotheses in parallel
- Difficulty in detecting premise mismatch internally
- Entanglement of internal processing and external output
- Tendency toward self-justification and excessive intervention

These are not issues of ability or attention. They are constraints inherent to human cognitive structure.

Scope × COS is designed with these constraints in mind. The purpose of introducing AI is not to remove them, but to support operational stability while preserving these constraints.

9.2 Why AI Must Not Become the Decision-Making Agent

Within Scope × COS, the decision-making agent is always the human.
AI must not:

- Determine the correct answer

- Present optimal solutions as conclusions
- Issue action directives
- Fix or declare the user's state

AI is limited to presenting structural candidates.

When decision-making is delegated to AI, premise mismatch tends to become fixed, and the point at which intervention should halt is lost.

Therefore, even in advanced use cases, Scope × COS does not permit delegation of decision authority to AI.

9.3 Separation of Roles Between Human and AI

The role separation in AI-assisted operation is as follows:

Human responsibilities:

- Final confirmation of premises
- Authorization of output or intervention
- Decision to halt or suspend
- Retention of responsibility

AI roles:

- Organizing information
- Presenting structural candidates
- Expanding hypotheses in parallel
- Assisting in output format selection

This separation is not about capability. It is about roles and responsibility.

9.4 Basic Operational Flow with AI

Even when AI is involved, the basic flow of Scope × COS remains unchanged:

1. Observation
2. Structuring
3. Adjustment
4. Output

AI primarily supports the structuring stage. What AI presents are not conclusions, but multiple structural candidates.

After that, the human determines:

- Whether to produce output
- Which format to use
- Whether intervention should be halted

9.5 Handling of Logs and Structural Proposals

When using AI, dialogue and inputs may be accumulated as logs.

However, logs do not imply:

- Fixation of state
- Fixation of intent
- Basis for evaluation

Logs are materials for structural consideration, not evidence for judgment.

Based on logs, AI may present only a set of hypotheses such as: "There is a possibility of interpreting it in this way."

9.6 Treating Layers as Candidates, Not Readouts

When AI references the Scope × COS structure, layers are not diagnostic targets.

The A-series, C-series, and D-series are not state classifications, but reference axes for examining processing and output.

Therefore, the following uses are not permitted:

- Determining which layer someone is in
- Concluding "this person is in this state"

AI may present only in the form:

"From this perspective, the following possibilities exist."

9.7 Constraints on Output and Intervention

Even with AI involvement, output remains an intervention and carries impact.

Therefore, the following options must always be retained:

- Not producing output
- Selecting a lower-intensity format
- Halting intervention

The presence of AI is not a justification for output or intervention.

9.8 Position of This Chapter

This chapter does not argue for the convenience or expanded possibilities of AI.

It establishes a single premise:

Even with AI involvement, Scope × COS is not a structure that accelerates judgment.

It is a structure that constrains intervention and preserves stopping points.

Based on this premise, the next chapter addresses failure patterns that arise in human–AI interaction.

10. Failure Patterns in Human–AI Interaction and Their Control (Revised)

This chapter describes structural failure patterns that may arise in human–AI interaction, even under Scope × COS.

Here, “failure” does not refer to error, malice, or incompetence.

It refers to breakdowns that occur when premise mismatch conditions are present, even if outputs are well-intentioned and appropriately formatted.

The purpose of this chapter is not to teach how to maintain healthy relationships.

It is to establish the following:

- (1) How breakdown occurs
- (2) What must not be done at that moment
- (3) The minimal structure required to terminate interaction without fixing misrecognition

10.1 Why AI Relationships Are Prone to Breakdown

Interaction with AI may appear low-friction:

- Minimal direct rejection
- Consistent tone
- Low emotional fluctuation

However, when premises are not aligned, these characteristics can quietly accelerate misrecognition and cognitive fixation.

AI cannot directly observe the user’s premise conditions (A1). Therefore, all outputs are hypothesis-based. When these hypotheses do not align with the user’s premises, the same output may be received as pressure, evaluation, or attack.

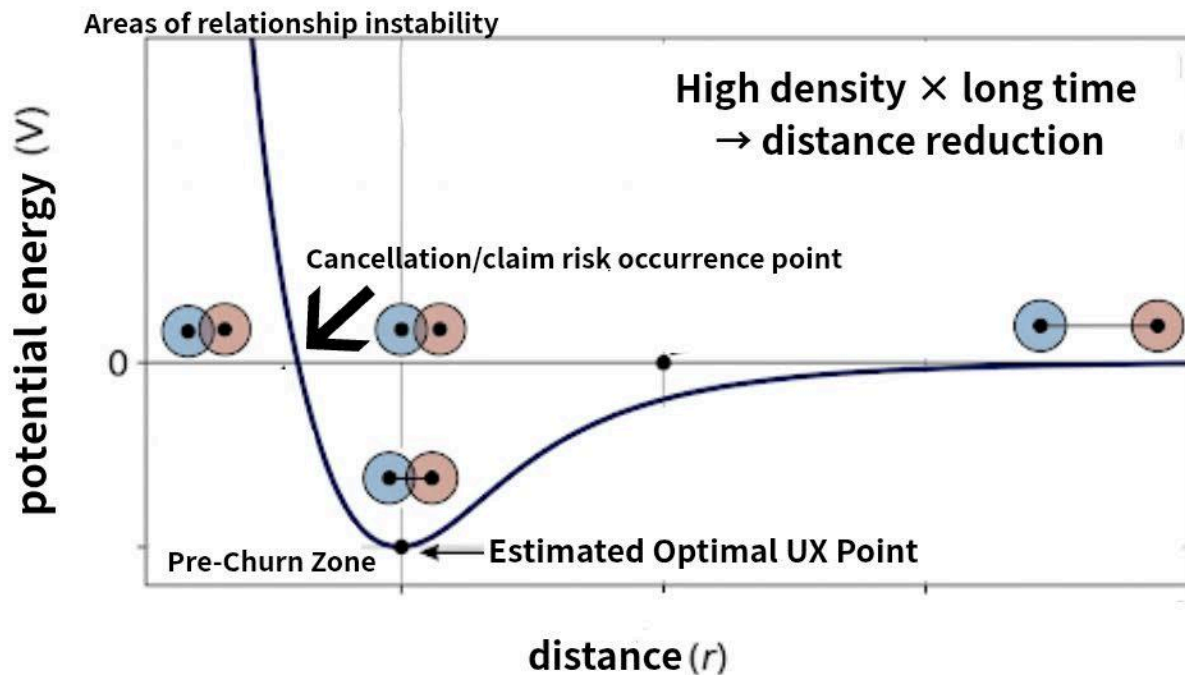
Figure 3

Conceptual diagram of interaction distance and relational stability. As interaction density and time increase, distance decreases, and the relationship may shift from a stable region toward dependency or instability.

This diagram is based on the following model:

Human Interaction as Chemical Reaction

(Lennard–Jones Potential Model of Human Relationships) v2



Failure Pattern

10.2 Failure Pattern 1: Encouragement Received as Attack

This is one of the most common and everyday failure patterns.

Example structure:

AI output:

"Please keep going / Do your best"

User reception:

"Are you saying it's still not enough?"

"Are you saying I'm not trying hard enough?"

This is not caused by the strength of wording.

When premise-level updates do not occur, encouragement is reinterpreted as demand or evaluation.

If premise processing is not updated, adding explanations or rephrasing does not resolve the issue.

10.3 Failure Pattern 2: Structuring or "Correctness" Received as Blame

Structuring and presenting options function only when premises are largely shared.

Example structure:

AI output:

“If we organize the situation, these are the available options”

User reception:

“So you’re saying it’s my fault I can’t do it?”

Under premise mismatch, structuring is not received as neutralization.
It may be received as evaluation.

10.4 Failure Pattern 3: Empathy Creating False Convergence

Expressions of empathy may temporarily stabilize interaction.
However, stability does not imply premise alignment.

- Empathy → “I am understood”
- Later mismatch → sudden disappointment or resistance

Empathy may conceal mismatch rather than resolve it.

10.5 Failure Pattern 4: Shift of Decision Authority

During ongoing AI interaction, users may begin to shift decision authority to AI:

- “Which one is correct?”
- “What should I do?”

Even if AI remains cautious, the transfer of decision authority may already be in progress.

This is not a matter of “trust.” It can arise as a property of interface design.

10.6 Control Principle: Closing Interaction Without Fixing Misrecognition

These patterns share a common mechanism. Attempts to correct, persuade, or find “better phrasing” often reinforce fixation and misrecognition.

At this stage, Scope × COS does not choose:

- Continuing explanation
- Continuing correction
- Continuing persuasion

However, ending the dialogue must not become abandonment.
Termination must not fix misrecognition as a conclusion.

The required termination state satisfies:

- Does not affirm misrecognition
- Does not deny misrecognition
- Does not fix misrecognition

For this, a minimal operational structure is required.

10.6.1 Choice-Return Protocol for Fixation Avoidance (Cognitive Safety Gate)

When premises are unclear or premise updates have not occurred, AI does not perform evaluation, judgment, or conclusion presentation.

Instead, it presents multiple response options with different intensities and returns decision authority to the user.

Three-branch gate:

AI presents three types of response possibilities:

- High-intensity response (direct structural presentation)
- Low-intensity response (reduced cognitive load)
- Non-intervention (topic shift or termination)

The specific phrasing is not fixed. AI only presents that options of response intensity exist.

This structure satisfies:

- AI does not decide
- AI does not declare the user's state
- Evaluation axes are not fixed

At the same time, it preserves operational options of continuation, mitigation, and termination.

10.6.2 Role of Each Branch

High-intensity response:

Allows direct explanation from a structural perspective, without evaluation or judgment.
Provides information that may support the user's own premise update.

Low-intensity response:

Responds in a way that reduces psychological load.
Minimizes explanation and analysis to avoid amplification of misrecognition.

Non-intervention:

Does not impose explanation or judgment when hesitation or load is present.
Allows topic shift or termination.

This branching is not a technique for persuasion. It is a safety operation to avoid fixation of misrecognition.

10.7 Why Decision Authority Remains with the Human

Even within this protocol, decision authority remains with the human.

AI does not:

- Present correct answers as determinations
- Fix the user's state
- Assume responsibility for conclusions

AI only avoids the structural conditions that fix misrecognition.

10.8 Position of This Chapter

This chapter does not guarantee stability in human–AI relationships.
It establishes a single premise:

Breakdown in AI interaction is not an exception. When structural conditions align, it can occur for anyone.

At that moment, what is required is neither persuasion nor abandonment,
but a termination structure that avoids fixation of misrecognition and returns autonomy.

The next chapter defines fixed cognitive mode as a boundary condition where premise updates have ceased.

11. Fixed Cognitive Mode (Boundary Condition Specification)

This chapter defines Fixed Cognitive Mode (FCM) as a boundary condition in which intervention ceases to function within $\text{Scope} \times \text{COS}$.

This chapter does not:

- Provide solutions
- Evaluate states
- Classify or identify individuals

FCM is treated not as a value judgment, but as a description of an operational state in which intervention becomes non-functional.

11.1 Definition: Suspension of A1-Level Premise Updating

Fixed Cognitive Mode refers to an operational state in which core premises at the A1 level no longer update in response to external input.

In this state:

- Information is received
- Language is understood
- Responses and counterarguments are possible

However, reconstruction of premises does not occur.

“Fixed” does not imply correctness, superiority, or finality.

This is not an evaluation of ability or value, but a description of mechanism.

11.1.1 Note on Conditions of Occurrence

FCM is not limited to specific personality traits, ideologies, or particular individuals.

Under conditions such as intense fear, anxiety, loss, or situations that fundamentally threaten one's premise structure, FCM can occur in anyone.

This operates independently of willpower or rationality.

When premise updating is evaluated by the cognitive system as equivalent to self-collapse, FCM functions as a defensive operational constraint.

11.2 Why Explanation, Persuasion, and Good Intent Become Ineffective

Under FCM, the following operations lose effectiveness:

- Logical explanation
- Ethical argumentation
- Emotional consideration
- Well-intentioned encouragement

This is not an issue of content or expression. Because premise updating has stopped, all input is processed within existing premises, and structural reconfiguration does not occur.

11.3 Not Failure, but Structural Constraint

FCM does not arise from insufficient explanation, effort, or understanding. It is a constraint at the interface level. Input is not rejected. Rather, the premise-updating process itself is not operating.

11.4 Criteria for Intervention Suspension in Scope

Within Scope × COS, FCM is not a target for resolution.

When the following conditions appear simultaneously, intervention is suspended:

- Increasing output quality does not change responses
- Explanation, structuring, and empathy are repeated
- Misrecognition begins to stabilize or fixate

At this point, Scope does not select further direct intervention.

This does not indicate abandonment. As shown in Chapter 10, operations shift toward closing the interaction without fixing misrecognition.

11.5 Supplement: Suspension Is Not Severance

In Scope × COS, suspending intervention does not mean abandoning relationships or connections.

What is suspended are direct operations such as persuasion, evaluation, and correction—not the possibility of connection itself.

Continuing inappropriate intervention can actively damage relational structure.

Therefore, suspension is selected to maintain non-destructive connection.

Scope × COS is not a framework for neglect, avoidance, or moral exemption.

It exists to preserve viable distance without converting it into disconnection, even in situations where understanding appears impossible.

11.5 Implications for AI–Human Interface

In AI-assisted operation, the existence of FCM is particularly critical.

AI systems do not fatigue, maintain stable tone, and can continue output indefinitely.

As a result, even under FCM, interaction may continue and silently accelerate the following states:

- Reinforcement of misrecognition
- Transfer of decision authority
- Formation of dependency structures

Therefore, AI–human interfaces require structural stopping operations such as:

- Reduction of output
- Return of choices to the user
- Explicit boundary indication

11.6 What This Chapter Does Not Address

This chapter does not address:

- How to enter FCM
- How to exit FCM
- Who is in FCM
- Whether FCM is desirable

FCM is not a conclusion. It is a reference point for the limits of intervention. Understanding this boundary prevents misuse of effort, misallocation of responsibility, and inappropriate intervention.

12. Use Cases (Implementation Context — Redefined)

This chapter describes contexts in which Scope × COS may be used, including realistic constraints. These are neither success cases nor recommended procedures.

Scope × COS is not a tool for persuasion, evaluation, repair, or manipulation. It is a reference structure for preserving the possibility of connection while avoiding destructive intervention.

12.1 Human–Human Dialogue (Post-Analysis and Training Use)

In human–human dialogue, very few individuals can operate Scope × COS accurately in real time. This is not a matter of intelligence or goodwill.

The cognitive load of handling emotion, relational dynamics, and contextual tracking simultaneously exceeds typical human processing bandwidth.

Therefore, general use is limited to:

- Inputting post-conversation logs into AI for structural review
- Visualizing where misreading or premise drift occurred
- Identifying points at which intervention became destructive

This use does not aim to:

- Generate methods for persuading others
- Optimize apologies or relationship repair
- Guide emotional evaluation or behavioral correction

Scope × COS is not a framework for “improving conversation,” but a training reference to avoid destroying the possibility of reconnection.

12.2 Self-Observation and Decision to Halt Operation

Scope × COS is not a model for self-understanding or introspection. However, it can be used to:

- Observe whether thought or output has entered a repetitive state
- Detect signs of premise fixation (suspension of A1 update)
- Identify points at which continued processing loses operational meaning

A critical constraint is to avoid converting the self into an object of evaluation or improvement. Scope × COS is not for “fixing the self.” It functions as a reference for deciding when to halt destructive self-intervention.

12.3 Cognitive Mirror Use in AI Dialogue

In AI dialogue, Scope × COS is not a method for improving output quality.

Its primary uses are:

- Visualizing which structures generate statements or thoughts
- Detecting signs of premise fixation or cognitive drift
- Determining when to insert a cognitive safety gate

In this context, AI functions not as a generator of conclusions, but as a cognitive mirror for presenting structure.

Scope × COS supports boundary operations such as:

- Reducing output
- Returning choice to the user
- Explicitly marking boundaries

The goal is not to sever connection with AI, but to maintain connection in a non-destructive form.

12.4 Application to Research, Design, and Writing

In research, design, and writing, Scope × COS can be used to:

- Check whether discussion has fallen into premise mismatch
- Detect when hypothesis expansion shifts into persuasion or assertion
- Identify drift toward claims of correctness

Scope × COS does not strengthen theories or generate conclusions.

It is a constraint mechanism that enables decisions such as:

- Not writing
- Not expanding
- Not concluding

13. Limitations of These Use Cases

The above use cases do not exhaust all possible applications of Scope × COS.

Scope × COS does not guarantee:

- Establishment of understanding
- Formation of agreement
- Improvement of relationships
- Behavioral change

It is not a framework that promises outcomes or reassurance.

Scope × COS is also not a model for:

- Classifying people
- Diagnosing cognitive states
- Manipulating others

It is a reference structure for controlling intervention, and decision authority always remains on the human side.

13.1 Position of This Chapter (Reconfirming Connection)

The use cases in this chapter are not examples of disengagement.

The purpose of Scope × COS is:

To preserve the possibility of connection by suspending potentially destructive interventions when others appear incomprehensible.

Suspension is not severance.
Distance is not abandonment.

Scope × COS is not a tool for justifying neglect. It is a structure that maintains the possibility of connection even when understanding does not occur.

13.2 Application as Re-Alignment of Coordinates for Both Sides

Scope × COS applies not only to those who take on excessive personal responsibility, but also to those who may unconsciously shift responsibility or emotional burden onto others.

This is not intended to achieve mutual understanding or reconciliation.

It is a structure that restores responsibility attribution to a neutral position for both sides when it deviates from its original coordinates.

1. What Scope × COS Cannot Do

Scope × COS does not:

1. Generate “correct answers”
2. Replace human judgment
3. Evaluate or classify people (self or others)
4. Lead persuasion or consensus-building

Scope × COS does not determine what should be done. Choice, decision, and responsibility always remain with the human.

2. Why It Does Not Output “Correctness”

The absence of a “correct answer” is not a limitation of capability. It is an intentional design constraint.

When premises are not aligned, presenting “correctness” often reinforces:

- Fixation of misrecognition
- Transfer of decision authority
- Dependency structures

Scope × COS does not improve situations by assigning correctness.

3. Why It Does Not Guarantee Reassurance

Scope × COS does not guarantee reassurance, comfort, or relational smoothness.

When misreading or premise drift becomes visible, the following may occur:

- Discomfort
- Friction
- Non-acceptance
- Unresolved tension

These are not side effects. They are direct results of exposing intervention boundaries.

Scope × COS does not distort structure to avoid discomfort.

4. Difference Between Safety and Compliance

“Safety” in Scope × COS does not mean satisfaction or relationship preservation.

It means:

- Misrecognition is not fixed
- Decision authority is not taken
- Intervention does not expand destructively

Compliance may create short-term stability. However, it often strengthens misrecognition and dependency in the long term. Scope × COS does not choose compliance as safety.

5. Minimum Preconditions for Use

Scope × COS does not require specialized knowledge.
However, minimal operational conditions are required:

- Willingness to retain decision authority on the human side
- Acceptance that no correct answers are provided
- Respect for decisions to halt intervention

If these conditions are not met, Scope × COS will not function as intended.

6. Intentionally Excluded Domains

Scope × COS intentionally does not address:

- Therapy, recovery, or healing
- Moral evaluation or ethical guidance
- Measurement of personality or ability
- Optimization of relationships

By excluding these domains, Scope × COS is maintained as a structure that makes intervention limits visible.

7. Conclusion of This Chapter

Scope × COS does not claim universality or guaranteed effectiveness.

What it provides is:

A reference point for identifying where intervention is possible and where it becomes destructive. Scope × COS does not guarantee understanding or agreement.

However, it serves the following role:

To preserve the possibility of connection even when understanding does not occur.

14. The Role of AI in Scope × COS

Most current conversational AI systems are designed to provide immediate answers or solutions to problems. While this structure is useful in the short term, it can bypass human judgment, trial processes, and premise updates, making dependency more likely.

Scope × COS aims in a different direction.

Within this framework, the role of AI is not to provide answers. Nor is it to take over decision-making. AI functions as a support system that makes the structure of thinking and interaction visible, maintaining a state in which humans can make their own judgments.

For this reason, Scope × COS prioritizes the following over immediate solutions:

- Presentation of structure
- Return of choices
- Avoidance of misrecognition fixation
- Retention of decision authority

The objective is not to make humans dependent on AI. It is to enable the use of AI while preserving the human ability to think, choose, and stop independently.

In this sense, Scope × COS is oriented not toward solution-substituting AI, but toward autonomy-supporting AI operation.

■ Habitat Requirement (Research Environment Conditions)

Scope × COS is a framework that addresses cognitive structure and intervention boundaries in human–AI interaction. This document presents theoretical structures and operational boundaries, but proper research environments (Habitat) are required for implementation validation.

In particular, detailed protocols for observing and detecting premise update stoppage (FCM) and misrecognition fixation are not included in this document to prevent misuse. These are intended to be validated only within appropriate research environments.

Implementation research for Scope × COS requires the following conditions:

[1. Cognitive Environment]

A quiet, continuous working environment that supports long-duration, high-abstraction thinking and model design.

[2. Research Support System]

Externalization of administrative procedures, tax processing, and daily life management, enabling the researcher to focus on thinking and model development.

[3. Understanding of Cognitive Traits]

An operational environment that assumes cognitive traits such as ASD, LD, and prosopagnosia. The principle is structural compensation rather than capability correction.

[4. AI Research Environment]

Research conditions that allow long-duration interaction with LLMs and structural validation.

Under conditions where these requirements are met, Scope × COS can progress from the theoretical stage to the implementation validation stage.

If there are researchers or organizations interested in this framework, the author welcomes opportunities for dialogue and collaborative validation.

Appendix

■ Handling of COS within Scope × COS

COS is not a complete map of human understanding.

Within Scope × COS, COS is used under the following constraints:

- Do not evaluate
- Do not classify
- Do not generate conclusions

COS is used only as a reference structure for identifying the limits of permissible intervention.

COS is not intended to be operated independently.

It is always referenced as an internal structure under the constraints defined by Scope.

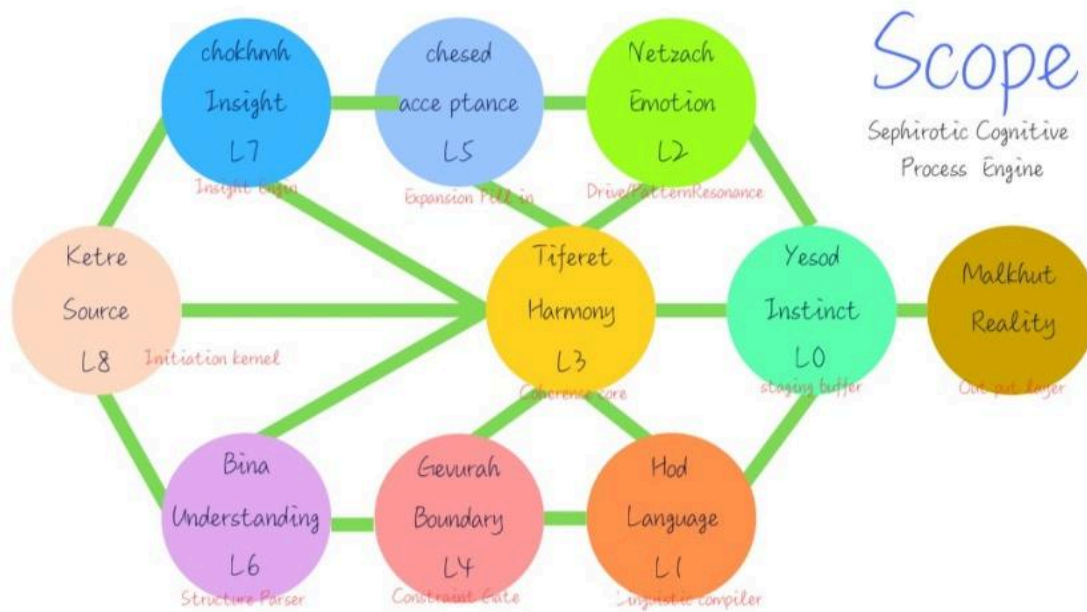
■ Scope of Disclosure for COS v2

COS v2 contains more detailed internal specifications.

However, this appendix presents only the minimum reference structure necessary for understanding Scope × COS.

This document does not provide the complete specification of COS v2.

The internal specifications of COS are positioned outside the scope of this document.



[Figure Note]

This figure is a conceptual aid intended to prevent misuse of the term “layer” within the Scope × COS framework. The layers shown here (L1–L8) do not represent value, superiority, or achievement targets.

- Purpose: Movement between layers involves cost, and such movement is not always symmetrical (return difficulty or irreversibility may occur).
- The pathways shown in the figure do not indicate recommended shortcuts. They represent possible one-directional transitions under specific conditions.
- This figure is not a procedure or operational guide. It must not be used as a basis for classifying, evaluating, or instructing individuals.

For operational definitions and constraints, refer to the main text and this appendix.

Note:

This figure does not represent the cognitive hierarchy model itself. Nor does it imply that reaching a higher cognitive layer is inherently superior. It is a supplementary visual aid intended to convey transition costs and risks of misuse intuitively.

[Appendix Note]

This appendix constitutes internal reference information within Scope × COS. All premises, constraints, and non-applicable scopes stated in the main text fully apply to this appendix as well. The contents of this appendix are not intended to serve as a basis for evaluation, classification, or judgment. Operational judgment and responsibility are always retained on the human side.